



# Crowd-Aware Socially Compliant Robot Navigation via Deep Reinforcement Learning

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## Abstract

Navigating in crowd environments is challenging for mobile robots because not only the safety but also the comfort of surrounding pedestrians must be considered. In this work, a deep reinforcement learning framework is introduced for safe and socially compliant robot navigation. We propose a value network for robot decision-making that leverages spatial-temporal reasoning to comprehend crowd interactions. Based on the real-time speed of pedestrians, the hazardous areas that the robot needs to avoid are designed, and a reward function is formulated to guarantee the safety and comfort of pedestrians. Extensive simulation experiments validate that the developed framework outperforms the state-of-the-art methods in terms of success rate (up to 34% increase) and discomfort frequency (up to 54.72% decrease). In addition, real-world experiments illustrate that our approach can predict pedestrian dynamics and navigate the robot safely and reliably in crowds.

**Keywords** Robot navigation · Motion planning · Deep reinforcement learning · Collision avoidance

## 1 Introduction

Robot navigation in crowded places has grabbed much attention in recent years [1–4]. Mobile robots are not only regarded as transportation media in social spaces, but also utilized to perform other services such as domestic cleaning and autonomous guidance. This requires robots to navigate safely and reliably in crowds, which is challenging due to the following two aspects. First, the high dynamics and

high stochasticity of crowd behavior threaten the security of human-robot interaction. Second, robots should abide by human social norms to ensure the comfort of pedestrians. Despite the challenges, robot social navigation has been extensively investigated, and many valuable achievements have emerged. Relevant research methods can be divided into reaction-based methods, trajectory-based methods and learning-based methods.

Among reaction-based methods, a pioneering work is the Social Force Model (SFM) [5], which uses attractive and repulsive forces to model crowd interactions. Other approaches such as Reciprocal Velocity Obstacle (RVO) [6] and Optimal Reciprocal Collision Avoidance (ORCA) [7] regard surrounding agents as velocity obstacles to calculate optimal collision-free velocities under reciprocal assumptions. The notable drawback of the above algorithms is that they rely heavily on hand-crafted rules. These methods exhibit effective obstacle avoidance in specific scenarios that meet the assumptions. However, in real social scenarios, the assumptions of these methods are often not fully satisfied, resulting in the effect of the algorithms being restricted, thus directly threatening the safety of human-robot interaction. Additionally, these methods suffer from the freezing robot problem in dense crowd environments [8, 9].

Trajectory-based methods learn human interaction behavior from pedestrian trajectory data [10, 11]. Kretschmar et

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al. [12] use maximum entropy inverse reinforcement learning (MaxEnt IRL) to imitate human behavior and rely on a Voronoi graph of the environment to plan paths. Che et al. [13] present a planning framework combining explicit and implicit communication and leverage MaxEnt IRL to obtain navigation policies. Trajectory-based methods predict pedestrian trajectories and allow robots to make far-sighted decisions. However, these approaches have the following drawbacks. First, it is computationally expensive for a robot to predict the trajectories of surrounding agents online and search for feasible paths from a vast navigation space. Second, predicted trajectories may render large portions of space unnavigable, and robots tend to behave too conservatively in response to such situations. Finally, these methods pay more attention to the safety of human-robot interaction, while neglecting the key element of human comfort.

Recently, learning-based methods are widely concerned. Several studies utilize imitation learning to uncover navigation strategies from demonstrations of desired behaviors. In [14–16], navigation policies with depth images, lidar measurements and local maps as input are developed by imitating expert demonstrations. Another line of works treat robot social navigation as a Markov decision process and study it using deep reinforcement learning, where human safety and comfort are taken into account [17, 18]. Chen et al. [19] present a decentralized multi-agent collision avoidance algorithm, which develops a value network that encodes the estimated time to the goal based on the joint state of the robot and its surrounding agents. Subsequently, Everett et al. [20] put forward an improved algorithm that adopts Long Short-Term Memory (LSTM) to sequentially process the states of adjacent agents, allowing the robot to navigate among an arbitrary number of agents. However, the network models of the above methods are relatively simplified and difficult to capture the potential impact of the crowd, which is not conducive to safe human-robot interaction. In contrast to these simplifications, Chen et al. [21] propose a value network that jointly models human-robot as well as human-human interactions to improve the robot's decision-making capabilities. Nevertheless, this method captures the spatial relationship of pedestrians while ignoring the temporal relationship of their movements, which affects the robot's ability to predict pedestrian trajectories. Moreover, in their simulation environment, the maximum speed of simulated humans is set to  $1m/s$ , while according to the relevant research on human gait, the normal walking speed of pedestrians in real scenes can reach up to  $2.5m/s$  [22], which will influence the method generalization ability in the real world and threaten human security. The latest achievement [23] models danger zones for the robot based on human velocity, enabling safe human-robot interaction. However, this approach does not adequately ensure the comfort of pedestrians. Based on human speed and human-robot proxemics [24], we respec-

tively devise the human action space and human personal space that the robot needs to avoid, which greatly pledges the safety and comfort of humans.

Through the above discussion, although the existing methods have achieved satisfactory results in robot social navigation, there is still much room for improvement. In this paper, we further investigate robot social navigation in crowded environments using deep reinforcement learning. We name our approach *spatial-temporal reasoning for crowd-aware reinforcement learning* (ST-CARL). The main contributions are summarized in the following three aspects. (1) A value network is proposed for robot decision making that incorporates spatial and temporal reasoning to understand crowd interactions. (2) Based on human speed and the comfortable distance allowed by humans, the hazardous areas for the robot are designed, and a reward function is formulated to fully guarantee the safety and comfort of pedestrians. (3) By means of the Robot Operating System (ROS), a robot social navigation system with our approach as the core is built, which can be well applied to real-world crowd scenarios.

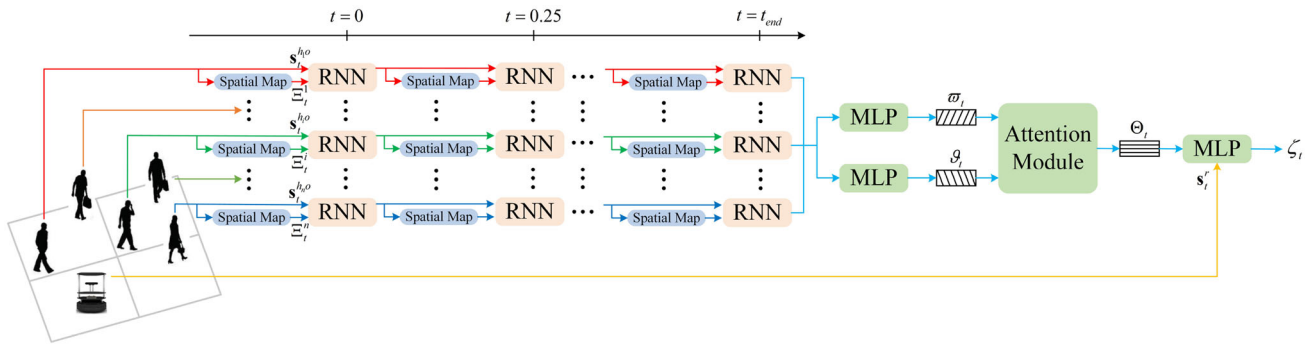
The rest of this paper is organized as follows. The description of our problem is introduced in Sect. 2. In Sect. 3, our methodology is described in detail. Section 4 presents extensive simulation experiments and real-world experiments to demonstrate the validity and practicality of the proposed approach. In Sect. 5, the limitations of our work are discussed and future work is provided. Eventually, conclusions are drawn in Sect. 6.

## 2 Problem Formulation

Consider the task of a robot navigating among  $n$  humans in a socially compliant manner and reaching the goal without collisions. This task can be expressed as a partially observable Markov decision process. The states of an agent (robot or human) are classified into observable states and unobservable states. The observable states are position  $\mathbf{p} = [p_x, p_y]$ , velocity  $\mathbf{v} = [v_x, v_y]$  and radius  $r$ , while the unobservable states are composed of goal  $\mathbf{g} = [g_x, g_y]$ , preferred speed  $v_{pref}$  and heading angle  $\theta$ . The robot-centric frame proposed in [19] is employed to make the state representation more general, where the origin is located at the robot position  $\mathbf{p}_t$ , and the x-axis points to the goal position  $\mathbf{g}$ . After parameterization, the robot state and the  $i$ -th human observable state at time  $t$  are represented as

$$\mathbf{s}_t^r = [d_g, v_x, v_y, r, v_{pref}, \theta], \quad (1)$$

$$\mathbf{s}_t^{hio} = [p_x^i, p_y^i, v_x^i, v_y^i, r^{h_i}, d^i, r^{h_i} + r], \quad (2)$$



**Fig. 1** Architecture of the proposed value network for crowd interaction analysis and understanding

where  $d_g = \|\mathbf{p}_t - \mathbf{g}\|_2$  indicates the distance from the robot  $\mathbf{p}_t$  to the goal  $\mathbf{g}$ , and  $d^i = \|\mathbf{p}_t - \mathbf{p}_t^i\|_2$  denotes the distance from the robot  $\mathbf{p}_t$  to the  $i$ -th human  $\mathbf{p}_t^i$ . At each time step, the robot can acquire its own state and the observable states of other agents. Consequently, the joint state at time  $t$  can be defined by combining the robot state and the observable states of all humans, namely  $\mathbf{s}_t^j = [\mathbf{s}_t^r, \mathbf{s}_t^{h1o}, \mathbf{s}_t^{h2o}, \dots, \mathbf{s}_t^{hno}]$ .

It is assumed that the robot can adjust its speed in time according to the action commands determined by the navigation policy at each time step, i.e.,  $\mathbf{v}_t = \mathbf{a}_t = \pi(\mathbf{s}_t^j)$ . Our aim is to find the optimal policy  $\pi^*(\mathbf{s}_t^j)$  to maximize the expected reward. Based on [21], the optimal value function is expressed as

$$V^*(\mathbf{s}_t^j) = \sum_{i=t}^T \gamma^{i-v_{pref}} R_i(\mathbf{s}_t^j, \pi^*(\mathbf{s}_t^j)), \quad (3)$$

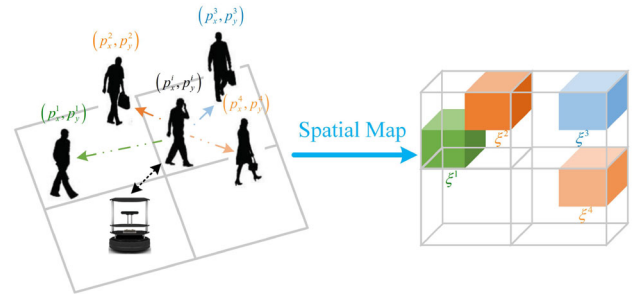
where  $\gamma \in (0, 1)$  is a discount factor. With the help of the value iteration method [25], the optimal policy is derived as

$$\pi^*(\mathbf{s}_t^j) = \arg \max_{\mathbf{a}_t \in \mathbf{A}} R(\mathbf{s}_t^j, \mathbf{a}_t) + \gamma^{\Delta t \cdot v_{pref}} \int_{\mathbf{s}_{t+\Delta t}^j} P(\mathbf{s}_{t+\Delta t}^j | \mathbf{s}_t^j, \mathbf{a}_t) V^*(\mathbf{s}_{t+\Delta t}^j) d\mathbf{s}_{t+\Delta t}^j, \quad (4)$$

where  $R(\mathbf{s}_t^j, \mathbf{a}_t)$  is the reward.  $\mathbf{A}$  is the action space, which includes the actions that the robot can perform.  $P(\mathbf{s}_{t+\Delta t}^j | \mathbf{s}_t^j, \mathbf{a}_t)$  is the transition probability from  $\mathbf{s}_t^j$  to  $\mathbf{s}_{t+\Delta t}^j$  when the action  $\mathbf{a}_t$  is implemented.

### 3 Methodology

In crowd interactions, humans adjust their paths by implicitly reasoning about the movements of their neighbors. In turn, these neighbors are influenced by others around them and may change their behavior over time. In this paper, we use spatial-temporal reasoning to capture the motion properties of pedestrians in crowd interactions. Besides, we devise the



**Fig. 2** Schematic diagram of spatial map for reasoning about the spatial relationship of pedestrians

hazardous areas that the robot needs to avoid to guarantee the safety and comfort of pedestrians in human-robot interaction. The details are described in the following subsections.

#### 3.1 Value Network

To better handle the high-dimensional state space, a deep neural network is used to approximate the value function that implicitly encodes social cooperation among agents. The proposed network architecture is shown in Fig. 1, where the spatial map and Recurrent Neural Network (RNN) modules capture the spatial and temporal relationships of pedestrians. Furthermore, the attention module is employed to calculate the relative importance of each pedestrian in the crowd.

The schematic diagram of the spatial map is displayed in Fig. 2. Given a neighborhood of size  $L$ , a map tensor  $\Xi_t^i$  of dimension  $L \times L \times 3$  centered on the  $i$ -th human is constructed to encode the existence and speed of surrounding humans

$$\Xi_t^i(m, n, :) = \sum_{j \in \mathcal{H}_t^i} \psi_{mn} [p_x^j - p_x^i, p_y^j - p_y^i] \xi^j, \quad (5)$$

where  $\mathcal{H}_t^i$  is the set of adjacent humans of the  $i$ -th human,  $\xi^j = (v_x^j, v_y^j, 1)$  denotes the state vector of the  $j$ -th human,  $\psi_{mn} [p_x^j - p_x^i, p_y^j - p_y^i]$  is an indicator function whose value

is equal to 1 only when the relative position  $(\Delta x, \Delta y)$  is situated in the cell  $(m, n)$ .

Subsequently, the  $i$ -th human observable state, the map tensor  $\Xi_t^i$  containing the spatial relationship of the  $i$ -th human, and the robot state are fed into the RNN

$$h_t^i = \text{RNN}(h_{t-1}^i, s_t^{hio}, \Xi_t^i, s_t^r; W_r), \quad (6)$$

where  $h_t^i$  is the hidden state of the RNN for the  $i$ -th human, and the RNN weights are denoted by  $W_r$ .

Since the number of pedestrians around the robot may vary considerably in different scenarios, the attention module is leveraged to process an arbitrary number of inputs into a fixed size output. The hidden state  $h_t^i$  is fed into a multi-layer perceptron (MLP) to acquire the interaction feature between the robot and the  $i$ -th human

$$\varpi_t^i = \Psi_{\varpi}(h_t^i; W_{\varpi}), \quad (7)$$

where  $\Psi_{\varpi}(\cdot)$  is a fully connected layer with ReLU nonlinearities and  $W_{\varpi}$  is the network weights. Meanwhile, the hidden state  $h_t^i$  is converted into an attention weight

$$\vartheta_t^i = \Psi_{\vartheta}(h_t^i; W_{\vartheta}), \quad (8)$$

where  $\Psi_{\vartheta}(\cdot)$  is a fully connected layer with ReLU activations and  $W_{\vartheta}$  is the network weights. After normalizing the attention weights, the crowd feature is the weighted sum of the interaction features

$$\rho_t^i = \frac{e^{\vartheta_t^i}}{\sum_{j=1}^n e^{\vartheta_t^j}}, \quad (9)$$

$$\Theta_t = \sum_{i=1}^n \rho_t^i \varpi_t^i. \quad (10)$$

Based on the crowd feature  $\Theta_t$  and the robot state  $s_t^r$ , the corresponding state value  $\zeta_t$  is estimated as

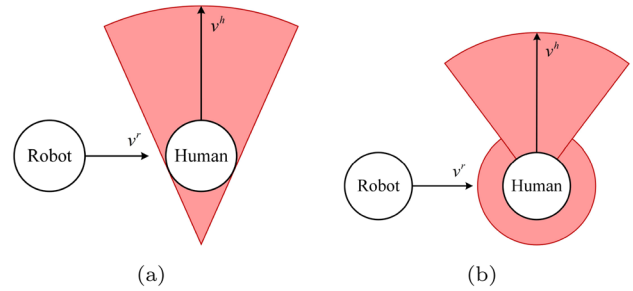
$$\zeta_t = \Psi_{\zeta}(\Theta_t, s_t^r; W_{\zeta}), \quad (11)$$

where  $\Psi_{\zeta}(\cdot)$  is a fully connected layer with ReLU nonlinearities and  $W_{\zeta}$  is the network weights.

By leveraging spatial-temporal reasoning to understand crowd interactions, the accuracy of state value estimation for densely populated scenarios is improved. In addition, based on the estimated state values, corresponding action instructions can be obtained to guide the robot to navigate safely and reliably in crowd environments.

### 3.2 Reward Function

In [23], the Danger-Zone is constructed for safe robot navigation, namely the fan-shaped area in Fig. 3a. However, in



**Fig. 3** **a** Illustration of danger-zone. **b** Illustration of the proposed hazardous area

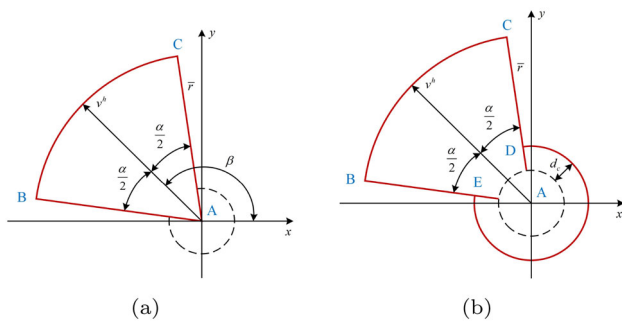
the formulation of the Danger-Zone, the authors focus more attention on the safety of the area in front of humans and ignore the impact on human safety and comfort when the robot is beside or behind them. This prompts us to redesign the hazardous area to fully pledge the safety and comfort of pedestrians in human-robot interaction.

Understanding human motion is critical for the robot to navigate in a safe and socially compliant manner. However, since the high stochasticity of human behavior, it is difficult for the robot to predict the next state of the human. A feasible solution is to consider the possible actions of the human in the current state to form an action space, and the robot is trained to avoid the above action space for safe and reliable navigation. Due to biological constraints, human locomotion is limited to certain actions. They cannot withstand large lateral deviations when moving at high speeds. Therefore, the human action space is closely related to the human velocity. When the speed is low, humans can move in any direction. As speed increases, humans move more forward and less sideways. In this paper, the initial geometry of the hazardous area is devised as a sector in the direction of human velocity to represent the human action space, that is, the region ABC in Fig. 4a, which is named the Risk-Zone (RZ). The circle represents the human and the point A is the center of the human. All angles are measured with the positive x-axis as the reference. Angles in the counterclockwise direction are set to positive, while angles in the clockwise direction are regarded as negative.  $\beta$  indicates the angle between the human velocity and the reference axis. For RZ, the radius and angle are determined by the human velocity

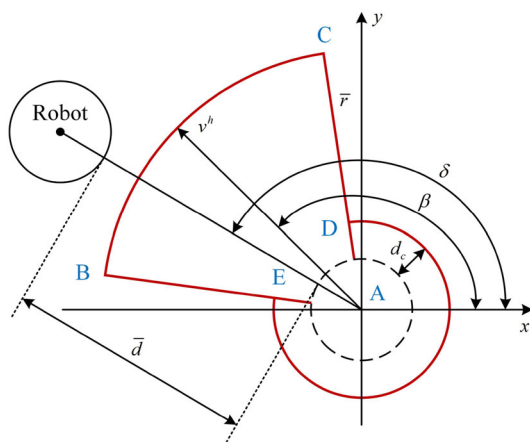
$$\bar{r} = w_v \cdot v^h + \tilde{r}, \quad (12)$$

$$\alpha = \frac{11\pi}{6} e^{-v^h} + \frac{\pi}{6}, \quad (13)$$

where  $w_v$  is the velocity weight,  $v^h$  is the human velocity,  $\tilde{r} = r^h + d_c$  is the sector radius when the human is stationary,  $r^h$  is the human radius, and  $d_c$  indicates the minimum comfortable distance accepted by the human. Obviously, as the human



**Fig. 4** **a** Initial geometry of hazardous area in the direction of human velocity. **b** Final geometry of hazardous area



**Fig. 5** Illustration of judging whether the robot is in the hazardous area

velocity  $v^h$  increases, the radius  $\bar{r}$  increases linearly and the angle  $\alpha$  decreases exponentially, and vice versa.

In [24], the author puts forward the theory of personal space based on the research of human interaction psychology, which provides a quantitative standard for the comfortable distance between human-robot interaction. Based on [24], we design the human personal space to guarantee the comfort of pedestrians. On the basis of RZ, a circle with point A as the center and  $\bar{r}$  as the radius is drawn to intersect the region ABC at points D and E. As shown in Fig. 4b, the annular region DE denotes the human personal space and is named the Discomfort-Zone (DZ). At this point, the final geometry of the hazardous area for the robot is determined as region BCDE. Figure 3b exhibits the interaction between the robot and the human when our method is employed. Owing to the presence of the proposed hazardous area, the robot keeps a distance from the human during navigation to ensure human safety and comfort.

To determine whether the robot is in the RZ, two conditions need to be verified. As shown in Fig. 5, the first condition is the distance, expressed by  $\bar{d} \leq \bar{r} - r^h$ , where  $\bar{d}$  is the minimum separation distance between the robot and the human in the time interval  $\Delta t$ . The second condition is the angle,

### Algorithm 1 ST-CARL Training

```

1: Initialize value network  $N$  with demonstration  $D$ 
2: Initialize target value network  $\tilde{N} \leftarrow N$ 
3: Initialize experience reply memory  $E \leftarrow D$ 
4: for episode = 1, ...,  $M$  do
5:   Initialize joint state  $s_0^j$  randomly
6:   repeat
7:      $\mathbf{a}_t \leftarrow \begin{cases} \text{RandomAction}(), & \epsilon \\ \arg \max_{\mathbf{a}_t \in \mathbf{A}} R(s_t^j, \mathbf{a}_t) \\ + \gamma^{\Delta t \cdot v_{pref}} N(s_{t+\Delta t}^j), & 1 - \epsilon \end{cases}$ 
8:      $value \leftarrow R(s_t^j, \mathbf{a}_t) + \gamma^{\Delta t \cdot v_{pref}} \tilde{N}(s_{t+\Delta t}^j)$ 
9:      $state \leftarrow s_{t+\Delta t}^j$ 
10:    Enrich experience  $E \leftarrow (state, value)$ 
11:    Update value network  $N$  by gradient descent
12:    Update time  $t \leftarrow t + \Delta t$  and state  $s_t^j \leftarrow s_{t+\Delta t}^j$ 
13:  until terminal state is reached  $s_t^j = s_{t_{end}}^j$  or timeout  $t > t_{max}$ 
14:  if episode | TargetUpdateInterval then
15:    Update target value network  $\tilde{N} \leftarrow N$ 
16:  end if
17: end for
18: return  $N$ 

```

denoted as  $\beta - \frac{\alpha}{2} \leq \delta \leq \beta + \frac{\alpha}{2}$ , where  $\delta$  is the angle of the robot position with respect to the reference axis. Intuitively, the robot is not in the RZ because the distance criterion is not satisfied. Furthermore, to judge whether the robot is in the DZ, only one condition needs to be checked, i.e.,  $\bar{d} < d_c$  and the robot is not in the RZ.

According to the above description of RZ and DZ, the reward function advanced in this paper is shown as follows, which rewards the robot for completing navigation tasks while penalizing it for colliding with humans. Moreover, the robot is also punished when it is in the RZ or the DZ.

$$R_t(s_t^j, \mathbf{a}_t) = \begin{cases} 1, & \text{if } d_g = 0; \\ -0.1, & \text{else if } \bar{d} < 0; \\ \frac{0.1\bar{d}}{\bar{r} - r^h} - 0.1, & \text{else if RZ;} \\ \frac{0.1\bar{d}}{d_c} - 0.1, & \text{else if DZ;} \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Therefore, based on the devised hazardous area and the corresponding reward function, the robot can avoid pedestrians in advance and keep a certain distance from them in real human-robot interaction, thereby achieving safe and socially compliant navigation.

### 3.3 Implementation Details

With the help of the proposed reward function, the value network is trained by the temporal difference approach with experience replay and fixed target network techniques, as

outlined in Algorithm 1. The value network is first initialized with imitation learning using demonstration experience (line 1–3) and then updated with reinforcement learning in which the robot acquires experience from exploration and interaction (line 4–17). In line 7, the  $\epsilon$ -greedy strategy is used to select the action. Evidently, an action is chosen randomly with probability  $\epsilon$  and the optimal action is chosen with probability  $1 - \epsilon$ . Based on the value network, we select optimal actions by repeatedly maximizing the one-step lookahead value. This corresponds to choosing the action that makes the joint state have the highest value. Furthermore, following [21], the next state  $\mathbf{s}_{t+\Delta t}^j$  is obtained by querying the true value from the simulation environment rather than estimating it with a linear motion model [19], which alleviates the problem of system dynamics in the training phase.

In the implementation, the spatial map is a  $4m \times 4m$  grid centered on each human, with each cell having a side length of  $1m$ . The hidden state size of the RNN is set to 50, and the dimensions of the hidden layers in  $\Psi_{\pi}(\cdot)$ ,  $\Psi_{\theta}(\cdot)$  and  $\Psi_{\zeta}(\cdot)$  are (50, 50), (50, 50) and (150, 100, 100), respectively. The preferred speed  $v_{pref}$  of the robot is set to  $1m/s$ . In addition, other related parameters such as  $r$  and  $r^h$  are set to  $0.3m$ ,  $w_v$  is set to 0.25,  $d_c$  is set to  $0.2m$ ,  $\Delta t$  is set to  $0.25s$ , and  $t_{max}$  is set to  $25s$ . In the imitation learning phase, the ORCA policy is leveraged to collect 3000 episodes demonstration, and the model is trained for 50 epochs with a learning rate of 0.01. Subsequently, the model is updated 10000 episodes with reinforcement learning, where  $\epsilon$  decreases linearly from 0.5 to 0.1 in the first 4000 episodes and then remains constant at 0.1 until the end. Moreover, the learning rate and discount factor  $\gamma$  are set to 0.001 and 0.9, respectively.

Consider a robot with non-holonomic kinematic constraints, whose action space is constructed as

$$\mathbf{A} = \left\{ (v, \omega) \mid v \in \mathcal{V}, \omega \in \mathcal{W} \right\}, \quad (15)$$

where the  $(v, \omega)$  pair represents an action, and  $v$  and  $\omega$  are the linear velocity and angular velocity of the robot, respectively. Specifically, the action space consists of 45 discrete actions: 5 velocities spaced exponentially between  $(0, v_{pref})$  and 9 orientations spaced evenly between  $[-\frac{\pi}{4}, \frac{\pi}{4}]$ , namely

$$\mathcal{V} = \left\{ \frac{e^{\frac{i+1}{5}} - 1}{e - 1} \cdot v_{pref} \mid i = 0, 1, \dots, 4 \right\}, \quad (16)$$

$$\mathcal{W} = \left\{ -\frac{\pi}{4} + \frac{\pi}{16} \cdot i \mid i = 0, 1, \dots, 8 \right\}. \quad (17)$$

## 4 Experiments

### 4.1 Simulation Experiment Setup

The experiments are carried out in an  $8m \times 8m$  simulation environment, which contains an arbitrary number of humans. All humans are controlled by ORCA and move randomly in the simulation environment. To match the real-world human speed, the minimum and maximum speeds of the simulated humans are set to  $0.5m/s$  and  $2.5m/s$ , respectively. The initial position and goal position of the robot are  $(0, -3m)$  and  $(0, 3m)$ , respectively. In order to comprehensively evaluate the validity of the proposed approach, the experimental scenes are divided into invisible scenes and visible scenes. The invisible scene sets the robot invisible to humans, i.e., humans only react to other humans but not to the robot. This setting removes ambiguity and directly shows the robot's decision-making ability when avoiding pedestrians. The visible scene sets the robot to be visible to humans, which is used to test the robot's navigation performance in a more realistic situation where the robot and humans interact with each other.

Comparative experiments with different crowd densities are set up to demonstrate the effectiveness of our approach. The proposed method and the state-of-the-art baseline methods are evaluated with 500 random test cases each. Performance indicators include success rate, collision rate, time, distance and discomfort frequency. Success rate (SR) is the rate at which the robot reaches the goal without collision, and collision rate (CR) is the rate at which the robot collides with humans. Time ( $\mu T$ ) is the average navigation time (in seconds) required for the robot to reach the goal. Distance ( $\mu D$ ) is the average minimum separation distance (in meters) between robot and human during interaction. Discomfort frequency (DF) is the ratio of the duration of the distance between robot and human less than  $d_c$  to the total navigation time.

### 4.2 Quantitative Evaluation

#### 4.2.1 Comparison of Network Models

Four network models in CADRL [19], LSTM-RL [20], SARL [21] and LM-SARL [21] are compared with our value network in an invisible scenario with five humans to demonstrate the superiority of the proposed network model. To eliminate ambiguity, all network models use the reward function advanced in this paper. The comparison results are shown in Table 1.

According to Table 1, it is distinct that CADRL and LSTM-RL have lower success rates. This is because the maxmin technique utilized in CADRL considers only one pair of interactions and ignores the others. For LSTM-RL,

**Table 1** Results of different network models

| Methods | SR          | CR          | $\mu T$     | $\mu D$     | DF           |
|---------|-------------|-------------|-------------|-------------|--------------|
| CADRL   | 0.61        | 0.39        | <b>7.46</b> | 0.10        | 0.159        |
| LSTM-RL | 0.70        | 0.30        | 9.75        | 0.12        | 0.113        |
| SARL    | 0.92        | 0.08        | 7.54        | 0.14        | 0.089        |
| LM-SARL | 0.93        | 0.07        | 7.75        | 0.14        | 0.083        |
| Ours    | <b>0.95</b> | <b>0.05</b> | 8.18        | <b>0.14</b> | <b>0.072</b> |

Bold indicates that this item has the best performance in all comparisons

the network uses the LSTM to process the states of adjacent agents sequentially based on the distance between them and the robot, but ignores their motion trends. Since the robot has difficulty reasoning adequately about the influence of neighboring agents on itself, it makes wrong decisions and collides with humans. In contrast, SARL, LM-SARL and our model all exhibit better results, with the proposed approach achieving the highest success rate of 95%. Based on the experimental results of the three methods, it can be seen intuitively that the navigation time of our model is relatively longer, but its discomfort frequency is 19.1% and 13.25% lower than that of SARL and LM-SARL, respectively. This indicates that the proposed model can predict human dynamics and make reasonable decisions by analyzing the spatial and temporal relationships of pedestrians.

#### 4.2.2 Ablation Experiments of the Reward Function

In order to prove the effectiveness of the hazardous area designed based on the human action space and personal space in the proposed reward function, ablation experiments are carried out in an invisible scene with five humans. The experimental results are displayed in Table 2. On the basis of our reward function, the reward function without RZ and DZ is regarded as Ours w.o. RZ & DZ, the reward function without RZ is set as Ours w.o. RZ, and the reward function without DZ is defined as Ours w.o. DZ. It is evident that due to the absence of hazardous areas, Ours w.o. RZ & DZ behaves very aggressively and causes the highest collision rate and discomfort frequency. Compared with Ours w.o. RZ, our method has a 5% higher success rate. This illustrates that the devised human action space can enable the robot to predict the possible next states of humans and avoid them in advance, so as to realize safe human-robot interaction. In addition, compared with Ours w.o. DZ, the proposed approach has a 0.03m longer average minimum separation distance and a 32.71% lower discomfort frequency. This reveals that the designed human personal space can keep a certain social distance between the robot and humans and effectively pledge the comfort of humans.

**Table 2** Results of ablation experiments

| Methods           | SR          | CR          | $\mu T$     | $\mu D$     | DF           |
|-------------------|-------------|-------------|-------------|-------------|--------------|
| Ours w.o. RZ & DZ | 0.85        | 0.15        | <b>7.93</b> | 0.11        | 0.124        |
| Ours w.o. RZ      | 0.90        | 0.10        | 8.72        | 0.12        | 0.096        |
| Ours w.o. DZ      | 0.93        | 0.07        | 8.05        | 0.11        | 0.107        |
| Ours              | <b>0.95</b> | <b>0.05</b> | 8.18        | <b>0.14</b> | <b>0.072</b> |

Bold indicates that this item has the best performance in all comparisons

**Table 3** Results of different reward functions

| Methods     | SR          | CR          | $\mu T$     | $\mu D$     | DF           |
|-------------|-------------|-------------|-------------|-------------|--------------|
| Chen et al. | 0.89        | 0.11        | 9.07        | 0.12        | 0.109        |
| Danger-Zone | 0.93        | 0.07        | <b>8.10</b> | 0.13        | 0.093        |
| Ours        | <b>0.95</b> | <b>0.05</b> | 8.18        | <b>0.14</b> | <b>0.072</b> |

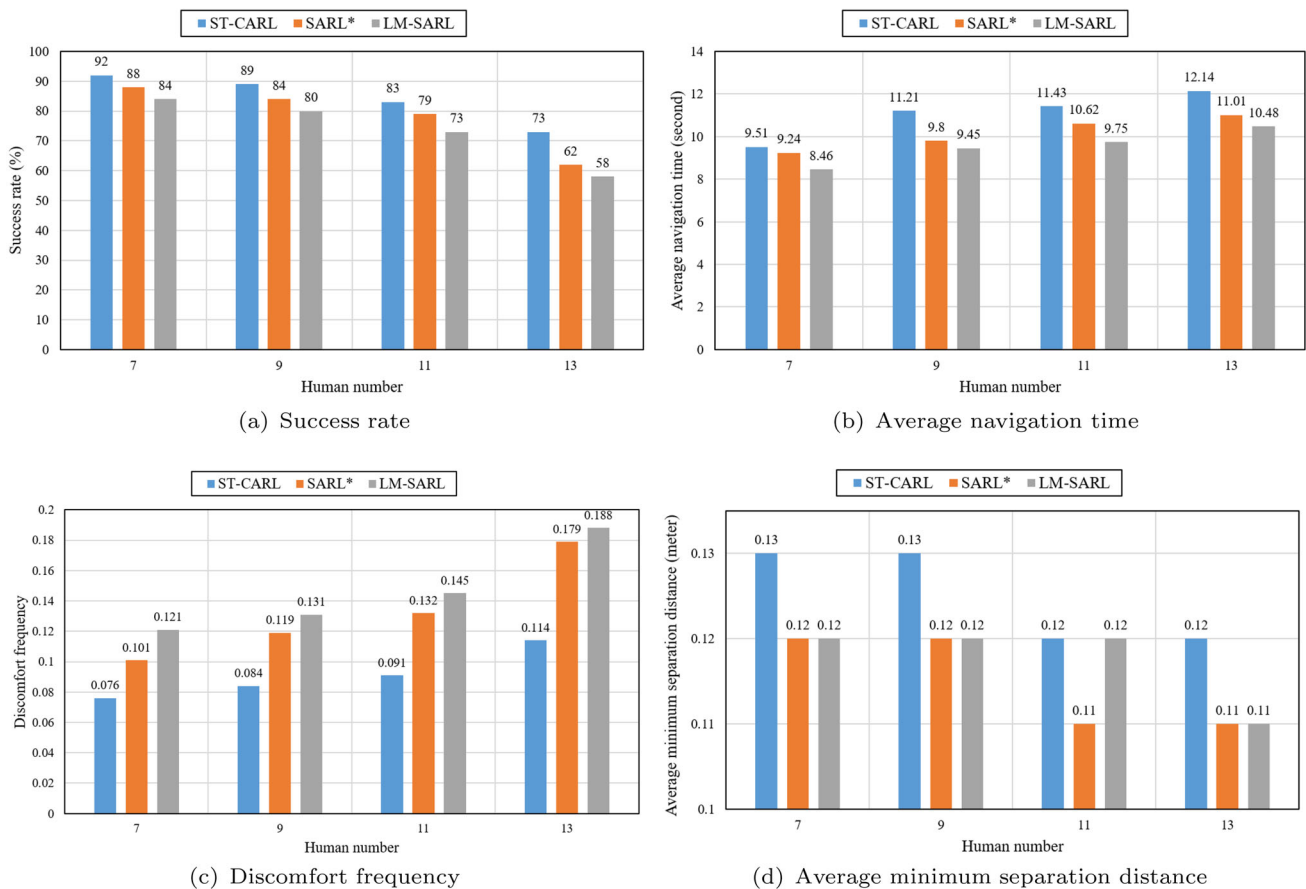
Bold indicates that this item has the best performance in all comparisons

#### 4.2.3 Comparison of Reward Functions

To further prove the validity of the proposed reward function, our reward function is contrasted with the latest reward functions Chen et al. [21] and Danger-Zone [23] in an invisible scenario involving five humans. Moreover, for fair comparison, the network model is fixed as the proposed value network. The corresponding results are listed in Table 3. It is obvious that our reward function outperforms the state-of-the-art methods in all the remaining metrics except for the time metric. The proposed approach achieves the highest success rate of 95%, which is 6% and 2% higher than the success rates of other reward functions, respectively. In addition, it is worth noting that the discomfort frequency of our reward function is reduced by 33.94% and 22.58% compared to that of Chen et al. and Danger-Zone, respectively. The above results illustrate that our approach can better ensure the safety and comfort of pedestrians in human-robot interaction compared to the latest methods.

#### 4.2.4 Invisible Scene Experiments with Different Crowd Densities

The invisible scene experiments with different crowd densities are completed, where the number of humans  $n \in \{7, 9, 11, 13\}$ . The proposed method ST-CARL is compared with two state-of-the-art approaches LM-SARL [21] and SARL\* [23], and the relevant results are depicted in Fig. 6. From Fig. 6a, we can confirm that ST-CARL exhibits the highest success rate regardless of the number of humans, while the performance of SARL\* and LM-SARL deteriorates with the increase of the crowd density. This is due to the fact that SARL\* and LM-SARL do not sufficiently consider the spatial and temporal relationships of humans and make inappropriate decisions, which leads to the failure of



**Fig. 6** Results of invisible scenes with different crowd densities

the navigation task. According to Fig. 6b–d, it is distinct that ST-CARL takes longer time to reach the target than the other methods. This is because ST-CARL pays more attention to human safety and comfort and keeps sufficient distance from humans while navigating, which results in a longer navigation path, but its discomfort frequency is substantially reduced. Among them, when the number of humans  $n = 13$ , the discomfort frequency of ST-CARL decreases the most compared with that of SARL\* and LM-SARL, by 36.31% and 39.36%, respectively. On the whole, the comparison results reveal the superiority and robustness of the proposed approach with good generalization ability in scenes with different crowd densities.

#### 4.2.5 Visible Scene Experiments with Different Crowd Densities

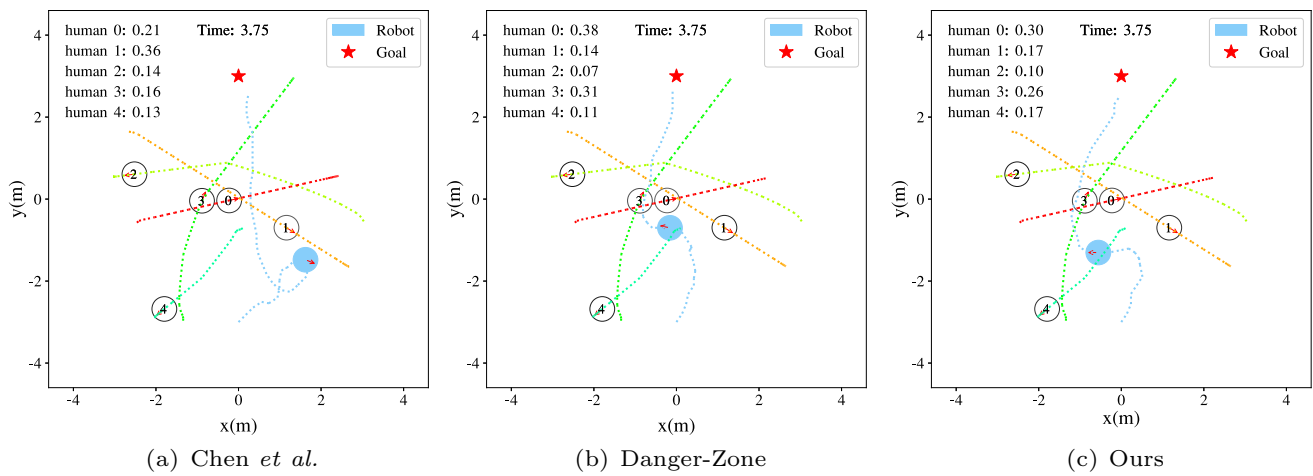
We further verify the effectiveness of the proposed approach in visible scenes with different crowd densities, where the number of humans  $n \in \{7, 9, 11, 13\}$ . The proposed ST-CARL is contrasted with LM-SARL [21] and SARL\* [23], and the related results are listed in Table 4. Comparing the results in Table 4 and Fig. 6a, it can be seen intuitively that

in visible scenes, the success rates of all methods are significantly improved, and the proposed approach shows the highest success rate in scenes with different crowd densities. This confirms that the proposed ST-CARL can guarantee the security of human–robot interaction better than the base-

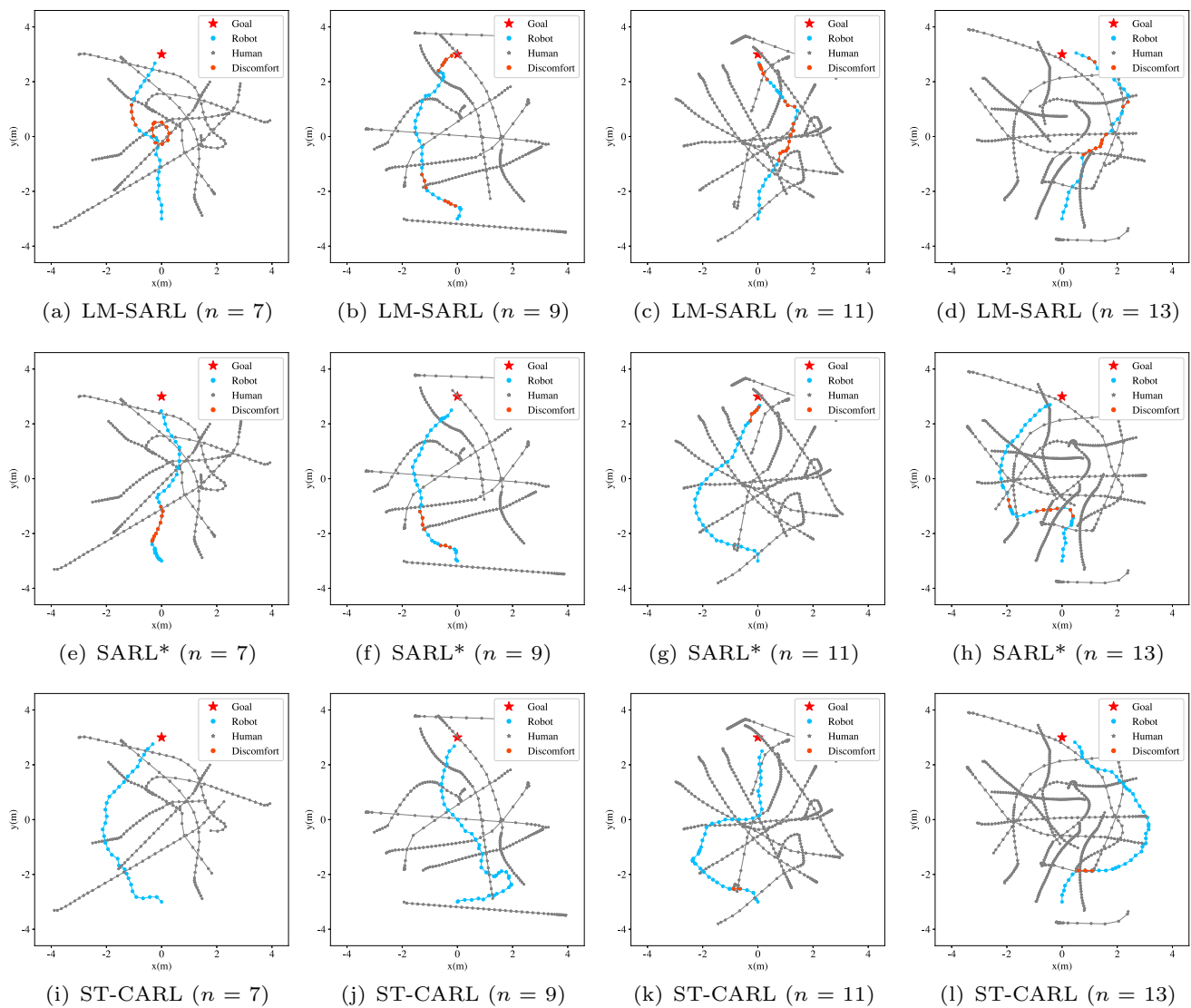
**Table 4** Results of visible scenes with different crowd densities

| Methods | $n$ | SR          | CR          | $\mu T$     | $\mu D$     | DF           |
|---------|-----|-------------|-------------|-------------|-------------|--------------|
| LM-SARL | 7   | 0.93        | 0.07        | 8.62        | 0.13        | 0.107        |
| SARL*   |     | 0.95        | 0.05        | <b>8.05</b> | 0.12        | 0.125        |
| ST-CARL |     | <b>0.98</b> | <b>0.02</b> | 9.29        | <b>0.13</b> | <b>0.073</b> |
| LM-SARL | 9   | 0.88        | 0.12        | 9.87        | 0.12        | 0.116        |
| SARL*   |     | 0.91        | 0.09        | <b>9.22</b> | 0.12        | 0.139        |
| ST-CARL |     | <b>0.95</b> | <b>0.05</b> | 10.72       | <b>0.13</b> | <b>0.087</b> |
| LM-SARL | 11  | 0.84        | 0.16        | 10.25       | 0.11        | 0.132        |
| SARL*   |     | 0.88        | 0.12        | <b>9.47</b> | 0.11        | 0.153        |
| ST-CARL |     | <b>0.93</b> | <b>0.07</b> | 11.37       | <b>0.13</b> | <b>0.089</b> |
| LM-SARL | 13  | 0.79        | 0.21        | 10.65       | 0.11        | 0.158        |
| SARL*   |     | 0.83        | 0.17        | <b>9.98</b> | 0.11        | 0.191        |
| ST-CARL |     | <b>0.88</b> | <b>0.12</b> | 11.90       | <b>0.12</b> | <b>0.118</b> |

Bold indicates that this item has the best performance in all comparisons



**Fig. 7** Navigation trajectories of different reward functions



**Fig. 8** Navigation trajectories of LM-SARL, SARL\* and ST-CARL for different crowd densities. Gray lines indicate human trajectories. The blue line represents the robot navigation trajectory, and the orange line denotes the robot navigation trajectory that causes human discomfort

line methods, regardless of the invisible scene or the visible scene. According to Table 4, it is worth noting that SARL\* reaches the target in the shortest time among the four sets of experiments, while its discomfort frequency is the highest compared to other methods. This is because SARL\* only considers pairwise interactions. When humans actively avoid the robot, SARL\* can make the robot behave more aggressively, which makes humans uncomfortable. In contrast, our method exhibits superior performance in discomfort frequency metric. Among them, when the number of humans  $n = 11$ , the discomfort frequency of the proposed method decreases the most compared with that of LM-SARL and SARL\*, by 32.58% and 41.83%, respectively. This demonstrates that the proposed approach can navigate in a socially compliant manner, effectively safeguarding human comfort.

### 4.3 Qualitative Evaluation

#### 4.3.1 Comparison of Reward Functions

In an invisible scene with five humans, the navigation trajectories of Chen et al. [21], Danger-Zone [23] and our reward function are displayed in Fig. 7. Meanwhile, Fig. 7 also reveals the positions of the robot and the humans at 3.75 s. In the formulation of Chen et al., the human motion is not considered, which prevents the robot from perceiving the risk posed by the rapid approach of humans in advance. It is evident from Fig. 7a that the robot detects the danger and turns around when human 1 is exceedingly close to the robot. This may significantly increase the probability of a collision between the robot and the human, thus endangering the safety and comfort of the human. On the other hand, the inability of the robot to make reasonable decisions in advance leads to a longer navigation trajectory, which affects operational efficiency and causes energy waste. According to Fig. 7b, it is distinct that the distance between the robot and human 0 is extremely small. As analyzed in Sect. 3.2, in the construction of Danger-Zone, there is no distance constraint on the robot in the human's side region, which easily leads to the robot being too close to the human during navigation, thus affecting the safety and comfort of the human. As can be seen in Fig. 7c, the proposed method overcomes the defects of the above reward functions and is able to maintain a safe distance from humans while navigating to ensure human safety and comfort.

#### 4.3.2 Invisible Scene Experiments with Different Crowd Densities

Figure 8 visualizes the navigation trajectories of LM-SARL [21], SARL\* [23] and ST-CARL for different crowd densities. As shown in Fig. 8a–d, since LM-SARL does not consider human motion, it is more aggressive during naviga-

**Table 5** Results of real-world experiments

| Methods | SR(%)      | $\mu T$ (second) | $\mu D$ (meter) | DF           |
|---------|------------|------------------|-----------------|--------------|
| LM-SARL | 70         | 29.3             | 0.15            | 0.171        |
| SARL*   | 90         | <b>26.6</b>      | 0.13            | 0.154        |
| ST-CARL | <b>100</b> | 28.5             | <b>0.16</b>     | <b>0.105</b> |

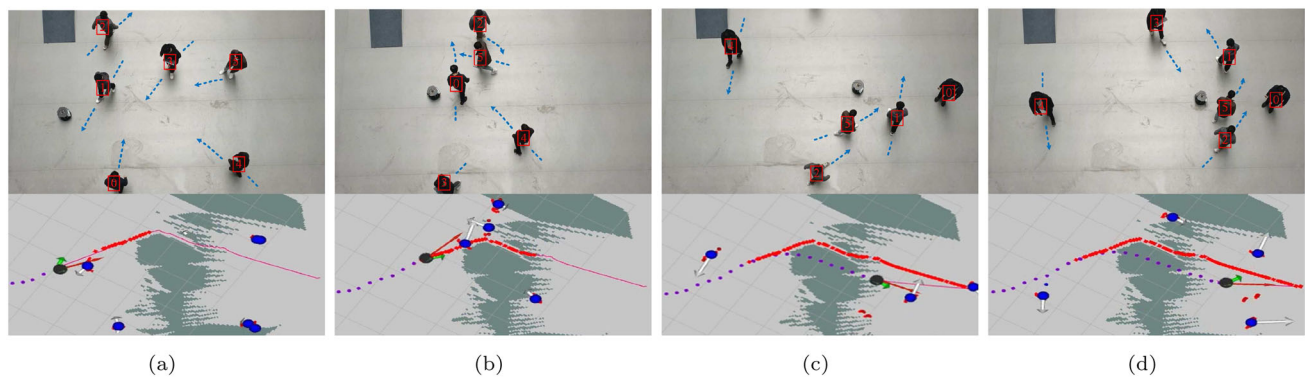
Bold indicates that this item has the best performance in all comparisons

tion and tends to move quickly through the crowd to reach the target, which greatly threatens human comfort and increases the risk of collision. It can be intuitively seen from Fig. 8e–h that the navigation trajectory of SARL\* is relatively conservative compared with that of LM-SARL. However, since the human comfort distance constraint is not considered in the formulation of Danger-Zone, the robot can arbitrarily enter the area that makes humans feel uncomfortable while navigating, thus affecting the human comfort. In the light of Fig. 8i–l, it is evident that since ST-CARL can jointly reason about the spatial and temporal relationships of humans, the robot is able to make appropriate decisions when interacting with the crowd. Besides, human safety and comfort are further guaranteed by benefiting from the design of the hazardous area in our reward function.

### 4.4 Real-World Experiments

In addition to the above simulation experiments, we also conduct navigation experiments in real crowd scenarios to verify the feasibility and validity of the proposed approach. A TurtleBot 2 mobile robot equipped with SLAMTEC RPLIDAR A3 is employed as the experimental platform. Moreover, the action commands for the robot are generated by the IRU-K10 minicomputer with Ubuntu 16.04 and ROS Kinetic installed. The map of the indoor experimental environment, the localization of the robot, and the position and velocity of the pedestrians are obtained through the ROS packages *gmapping*, *amcl* and *leg\_detector*, respectively. Furthermore, since the value of the time interval  $\Delta t$  is set small, we assume that the pedestrian speed from the current time step to the next time step is constant, and utilize the pedestrian position and speed at the current time step to estimate the pedestrian position at the next time step.

The real-world experimental environment is a rectangular field of  $8m \times 5m$ . The distance between the starting position and the target position of the robot is set to  $7m$ . There are six participants in the experiment, and they move randomly in the experimental environment according to their own intentions. In particular, participants are told to actively avoid the robot when they find that the robot is too close without avoiding them. And when this happens, the navigation task of the robot is counted as a failure. To demonstrate the advantage of the proposed approach, our method is contrasted with LM-SARL



**Fig. 9** Real-world robot social navigation among crowds

[21] and SARL\* [23]. Each method is run 10 times in the experimental environment. The relevant measures, namely success rate, average navigation time, average minimum separation distance and discomfort frequency, are recorded in Table 5. It is obvious that the proposed method successfully completes all navigation tasks. Moreover, the discomfort frequency of our method is 38.6% and 31.82% lower than that of LM-SARL and SARL\*, respectively. These results further confirm the superiority of our method, which effectively guarantees the safety and comfort of human-robot interaction.

The snippets of the real-world experiment of our method are exhibited in Fig. 9. The top half of the image is the panoramic view of the real-world experiment. The numbers in red are the participant numbers. The blue dashed lines with arrows indicate the trajectories of the participants. The bottom half of the image is the Rviz view, where the black circle represents the robot, the red long arrow and the green short arrow express the robot's movement direction and the action command to be executed by the robot, respectively. The blue circle and white arrow denote the position and velocity of the human, respectively, as measured by the 2D lidar. The purple dots indicate the trajectory that the robot has traveled, and the red dots are the local targets of the robot. It is apparent from Fig. 9 that the robot can perceive the surrounding pedestrians and comprehend their movement intentions, so that it can make reasonable decisions to avoid pedestrians and achieve safe and reliable social navigation.

## 5 Discussion

In this section, we state the limitations of our work and provide our future work accordingly. First, in simulation experiments, the simulated human motion is controlled by ORCA, which cannot fully fit the real human motion behavior. In future work, actual human trajectory data will be used for algorithm training and testing to further improve the robot's adaptability to real crowd scenarios. Second, in

real-world experiments, since only 2D lidar data is used, the detected human position and velocity are noisy or some pedestrians are not detected, which affects the robot's decision-making. In future work, multi-sensor information such as lidar data, RGB and depth images will be employed to improve pedestrian detection accuracy. Third, the research scenarios of this work do not contain static obstacles. In future work, socially compliant robot navigation in complicated obstacle environments will be further investigated.

## 6 Conclusions

This paper tackles the problem of socially compliant robot navigation in dense crowd scenarios with deep reinforcement learning. A novel value network is advanced, which integrates spatial and temporal reasoning for robot decision-making. Besides, based on human action space and personal space, the hazardous areas around humans are designed, and the robot is trained to avoid these hazardous areas for safe and credible navigation. Extensive simulation experiments prove that our approach outperforms the latest methods in the following metrics: high success rate and low discomfort frequency. Moreover, real-world experiments further verify the validity of our method, which can better guarantee human safety and comfort than state-of-the-art methods.

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## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

## References

- Kruse T, Pandey AK, Alami R, Kirsch A (2013) Human-aware robot navigation: a survey. *Robot Auton Syst* 61(12):1726–1743
- Hu Z, Zhao Y, Zhang S, Zhou L, Liu J (2022) Crowd-comfort robot navigation among dynamic environment based on social-stressed deep reinforcement learning. *Int J Soc Robot* 14(4):913–929
- Sathyamoorthy AJ, Patel U, Paul M, Kumar NKS, Savle Y, Manocha D (2022) CoMet: Modeling group cohesion for socially compliant robot navigation in crowded scenes. *IEEE Robot Autom Lett* 7(2):1008–1015
- Wang C, Chen X, Li C, Song R, Li Y, Meng MQ-H (2023) Chase and track: Toward safe and smooth trajectory planning for robotic navigation in dynamic environments. *IEEE Trans Ind Electron* 70(1):604–613
- Helbing D, Molnr P (1995) Social force model for pedestrian dynamics. *Phys Rev E* 51(5):4282–4286
- Van Den Berg J, Lin M, Manocha D (2008) Reciprocal velocity obstacles for real-time multi-agent navigation. In: 2008 IEEE International Conference on Robotics and Automation. IEEE, pp. 1928–1935
- Van Den Berg J, Guy SJ, Lin M, Manocha D (2011) Reciprocal n-body collision avoidance. *Robotics Research*. Springer, Berlin, pp 3–19
- Trautman P, Krause A (2010) Unfreezing the robot: Navigation in dense, interacting crowds. In: 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, pp. 797–803
- Sathyamoorthy AJ, Patel U, Guan T, Manocha D (2020) Frozone: freezing-free, pedestrian-friendly navigation in human crowds. *IEEE Robot Autom Lett* 5(3):4352–4359
- Kretzschmar H, Kuderer M, Burgard W (2014) Learning to predict trajectories of cooperatively navigating agents. In: 2014 IEEE International Conference on Robotics and Automation. IEEE, pp. 4015–4020
- Pfeiffer M, Schwesinger U, Sommer H, Galceran E, Siegwart R (2016) Predicting actions to act predictably: Cooperative partial motion planning with maximum entropy models. In: 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, pp. 2096–2101
- Kretzschmar H, Spies M, Sprunk C, Burgard W (2016) Socially compliant mobile robot navigation via inverse reinforcement learning. *Int J Robot Res* 35(11):1289–1307
- Che Y, Okamura AM, Sadigh D (2020) Efficient and trustworthy social navigation via explicit and implicit robot-human communication. *IEEE Trans Robot* 36(3):692–707
- Tai L, Zhang J, Liu M, Burgard W (2018) Socially compliant navigation through raw depth inputs with generative adversarial imitation learning. In: 2018 IEEE International Conference on Robotics and Automation. IEEE, pp. 1111–1117
- Long P, Liu W, Pan J (2017) Deep-learned collision avoidance policy for distributed multiagent navigation. *IEEE Robot Autom Lett* 2(2):656–663
- Liu Y, Xu A, Chen Z (2018) Map-based deep imitation learning for obstacle avoidance. In: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, pp. 8644–8649
- Chen YF, Everett M, Liu M, How JP (2017) Socially aware motion planning with deep reinforcement learning. In: 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, pp. 1343–1350
- Cui Y, Zhang H, Wang Y, Xiong R (2021) Learning world transition model for socially aware robot navigation. In: 2021 IEEE International Conference on Robotics and Automation. IEEE, pp. 9262–9268
- Chen YF, Liu M, Everett M, How JP (2017) Decentralized non-communicating multiagent collision avoidance with deep reinforcement learning. In: 2017 IEEE International Conference on Robotics and Automation. IEEE, pp. 285–292
- Everett M, Chen YF, How JP (2018) Motion planning among dynamic, decision-making agents with deep reinforcement learning. In: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, pp. 3052–3059
- Chen C, Liu Y, Kreiss S, Alahi A (2019) Crowd-robot interaction: Crowd-aware robot navigation with attention-based deep reinforcement learning. In: 2019 IEEE International Conference on Robotics and Automation. IEEE, pp. 6015–6022
- Bohannon RW (1997) Comfortable and maximum walking speed of adults aged 20–79 years: reference values and determinants. *Age Age* 26(1):15–19
- Samsani SS, Muhammad MS (2021) Socially compliant robot navigation in crowded environment by human behavior resemblance using deep reinforcement learning. *IEEE Robot Autom Lett* 6(3):5223–5230
- Hall ET (1966) *The hidden dimension*. Anchor, Garden City, NY
- Sutton RS, Barto AG (1998) *Reinforcement learning: An introduction*. MIT Press, Cambridge, MA

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